

Bonus task 7

Equivalent forces

Possibility to earn five bonus points

Deadline: Before the 21st of June

Your task is to compute the equivalent nodal forces for a four-node bilinear (Q4) and an eight-node bi-quadratic (Q8) plane element subjected to various types of loading. For each load case, derive the consistent nodal load vector

$$\mathbf{f}_{equiv}^{bf} = \int_V \mathbf{b}^T \mathbf{N}_m dV \quad (1)$$

$$\mathbf{f}_{equiv}^{tf} = \int_{\Gamma} \mathbf{q}^T \mathbf{N}_m d\Gamma \quad (2)$$

depending on whether the loading is a body force or a boundary (surface) traction.

Load Cases

1. **Uniform surface traction** Applied on the right edge:

$$\mathbf{t} = \begin{bmatrix} 0 \\ t_0 \end{bmatrix}.$$

2. **Linearly varying traction** Applied on the bottom edge:

$$\mathbf{t}(x) = \begin{bmatrix} 0 \\ t_0 \left(1 + \frac{x}{L}\right) \end{bmatrix}.$$

3. **Uniform body force** A constant vertical load such as gravity:

$$\mathbf{b} = \begin{bmatrix} 0 \\ -\rho g \end{bmatrix}.$$

4. **Hydrostatic pressure on a vertical edge** The pressure varies with height:

$$p(y) = \rho g(h - y),$$

leading to a boundary traction

$$\mathbf{t}(y) = -p(y) \mathbf{n}.$$

Instructions

For each loading case:

- Write the shape functions for both Q4 and Q8 elements.
- Form the boundary or volume integrals.
- Compute the Jacobian for edges (surface loads) or the full element (body loads).
- Perform numerical integration (Gaussian quadrature).
- Assemble the final consistent nodal load vector.

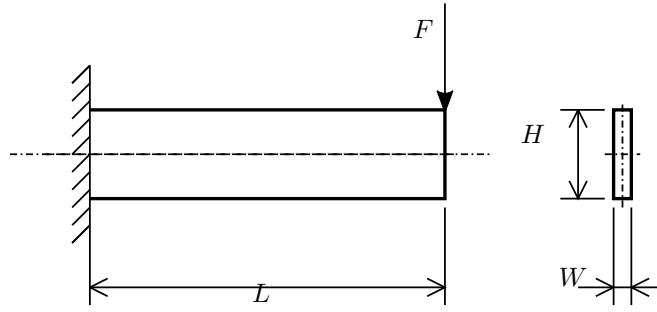


Figure 1. Thick cantilever problem

The goal is to understand how different load distributions map to equivalent nodal forces in isoparametric elements and how the order of interpolation (Q4 vs. Q8) affects the result.

The parameters of the problem are given in Table.

Force, F [N]	Young's mod., E [Pa]	Poisson's ratio, ν	Length, L [m]	Height, H [m]	Width, W [m]
62500	$2.07 \cdot 10^{11}$	0.3	2	0.5	0.1

Write a small report that presents global displacements (average) at the free end explicitly in the table with various meshes, and present convergence curves. Submit the report in PDF format. Also, submit all necessary MATLAB/Python files (m-files/py-files) and verify that they run without issues. Bonus points will be shared based on the MATLAB/Python solution as follows: correct derivation of the element with reduced integration scheme (1 point), solved displacements (average) in tables with both elements (1 point), convergence curves with both elements (2 points), and correct explanation for the difference between elements (1 point). Please note that no points without a clear report (PDF format!) and MATLAB/Python codes (m-files!).

Code

All the necessary codes to perform static analysis of the cantilever problem are provided to solve the problem. You need to make a small code that presents convergence curves.

run_DeriveQ4xi_FullGauss.m.py	run firstly to export stiffness matrix
run_DeriveQ4psR_v12.m/.py	modify numerical integration and run to export stiffness matrix
run_LinStatic_Q4_v20.m/.py	Solves cantilever problem
RectangularMeshQ4.m/.py	Makes list of nodal positions and element's nodal connectivity
ElemDOFsConnectivity_Q4.m/.py	Makes DOF's connectivity from nodal connectivity
DofID.m/.py	Returns DOF ID with given nodes and nodal DOFs

Convergence curves

To produce convergence curves, you can estimate vertical displacement error between FE solution and analytical solution from beam theory as $Error = |v_{FEM} - v_{max}|$. Present error in the y-axis and DOF's in the x-axis.

TIPS

To get rid of shear locking (of bilinear plate element), you should make separate integration for energies related to shear and normal deformations.

$$U = U_n + U_s \quad (3)$$

$$U_n = \frac{1}{2} l_z \int_V \underbrace{\boldsymbol{\sigma}_n^T \boldsymbol{\varepsilon}_n \det(\mathbf{J}_e)}_{f_n(\xi, \eta)} d\xi = \frac{1}{2} l_z \int_V \boldsymbol{\varepsilon}_n^T \mathbf{D}_n \boldsymbol{\varepsilon}_n \det(\mathbf{J}_e) d\xi \quad (4)$$

$$U_s = \frac{1}{2} l_z \int_V \underbrace{\sigma_{xy} \gamma_{xy} \det(\mathbf{J}_e)}_{f_s(\xi, \eta)} d\xi = \frac{1}{2} l_z \int_V D_s \gamma_{xy}^2 \det(\mathbf{J}_e) d\xi \quad (5)$$

Note that $f_n(\xi, \eta)$ and $f_s(\xi, \eta)$ are integrands, and the number of integration points is estimated based on those functions.

Analytic solution based on beam theory

$$\frac{d^2 v}{dx^2} = -\frac{M(x)}{EI(x)} - \frac{\kappa_{xy} q(x)}{GA} \quad (6)$$

Maximum deflection for the cantilever beam under applied force at the free end can be found as:

$$v_{max} = \underbrace{\frac{FL^3}{3EI}}_{\text{bending}} + \kappa_{xy} \underbrace{\frac{FL}{GA}}_{\text{shear}} \quad (7)$$

where κ_{xy} is a shear correction factor (needed to use in the beams based on the first-order shear deformation theory (FSDT)). It can be approximated to be 6/5 for a rectangular cross-section.